

Shrinkage Stoping

17/06/2020

Shrinkage Stoping 1

- *Shrinkage stoping is vertical or steeply Deeping, open, over hand mining method.
- *Broken Rock remain in stope to form plat form.
- *Broken rock provides additional support to HW.
- *Stopes are mined upwards.
- *Horizontal slices are drilled by hand held drill machines.
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Ore Body 1

- *Ore body should be vertical or steeply Deeping.
- *Ore body should be regular boundaries.
- *Ore should be fairly continuous along strike direction.
- *Broken ore should flow freely down wards on foot wall.
- *Ore body, Hanging Wall, Foot Wall should not contain clay or any other sticky material.
- *Broken ore should not have tendency of packing (clustering).

Ore Body 2

- *Ore should not be readily oxidized (Re cementing)
- *Oxidation of ore has adverse effect on ore processing.
- *This causes broken ore to re cemented it self & causes “hang-up”.
- *Ore should be free from waste.
- *Hang wall, foot wall should be competent, self supporting & stand intact till end of stoping.
- *Shrinkage method can not be applied to weak rock formation.
- *This method can be applied in ore body from 1.2m to 30m.

Stope Development 1

- *Cross cut to ward ore body from shaft.
- *Upper level & lower derive are excavated in side ore body (vein).
- *Drift at FW of the ore body is excavated in some options.
- *These two levels are kept 30m to 180m vertically apart (level deference).
- *Drives are excavated by manual drilling, blasting or with help of trackmounted machinery.
- *Sill level in the stope 3 to 5 m above drift.
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Stope Development 2

- *Normally two raises in ore is excavated from lower level to upper level for ventilation, access etc.
- *Raises are excavated in ore body.
- *Man Holes in rises are excavated to have access to stope.
- *Excavation of raises are done manually, by Alimak Raise climber or raise borer.
- *Ore Draw Raise are excavated through Sill Pillars.
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Stope Development 3

Extraction arrangement with raise.

- *Extraction raises are excavated in sill pillar 7.5m to 9m long.
- *Bottom of raise is fitted with chute gate.
- *Chute can be of timber construction or steel structure.
- *Upper portion of extraction raises are bailed out (hogged over) after under cut for first stope slice (sill level).
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Stope Development 4

Extraction arrangement with out raise.

- *Two horizontal slices are cut from bottom of stope ore block.
- *Broken ore is cleaned up from bottom of stope ore block.
- *Timber chute or steel chute gate are installed at parallel drive in waste rock at foot wall of orebody.
- *Theses chute gates are placed at interval of 7.5m interval.

Stope Development 5

Extraction cross cut & LHD.

- *Extraction cross cut is excavated parallel to ore body about 7.5m to 15m in foot wall of ore body.
- *Draw holes (extraction cross cut) are driven from foot wall drift to ore drift spaced at 7.5m to 15m.
- *Back of ore body (sill level) is blasted down & swell is extracted through draw holes with rail mounted loading machine or Load haul Dumping machines (LHD).

Shrinkage Stope With LHD

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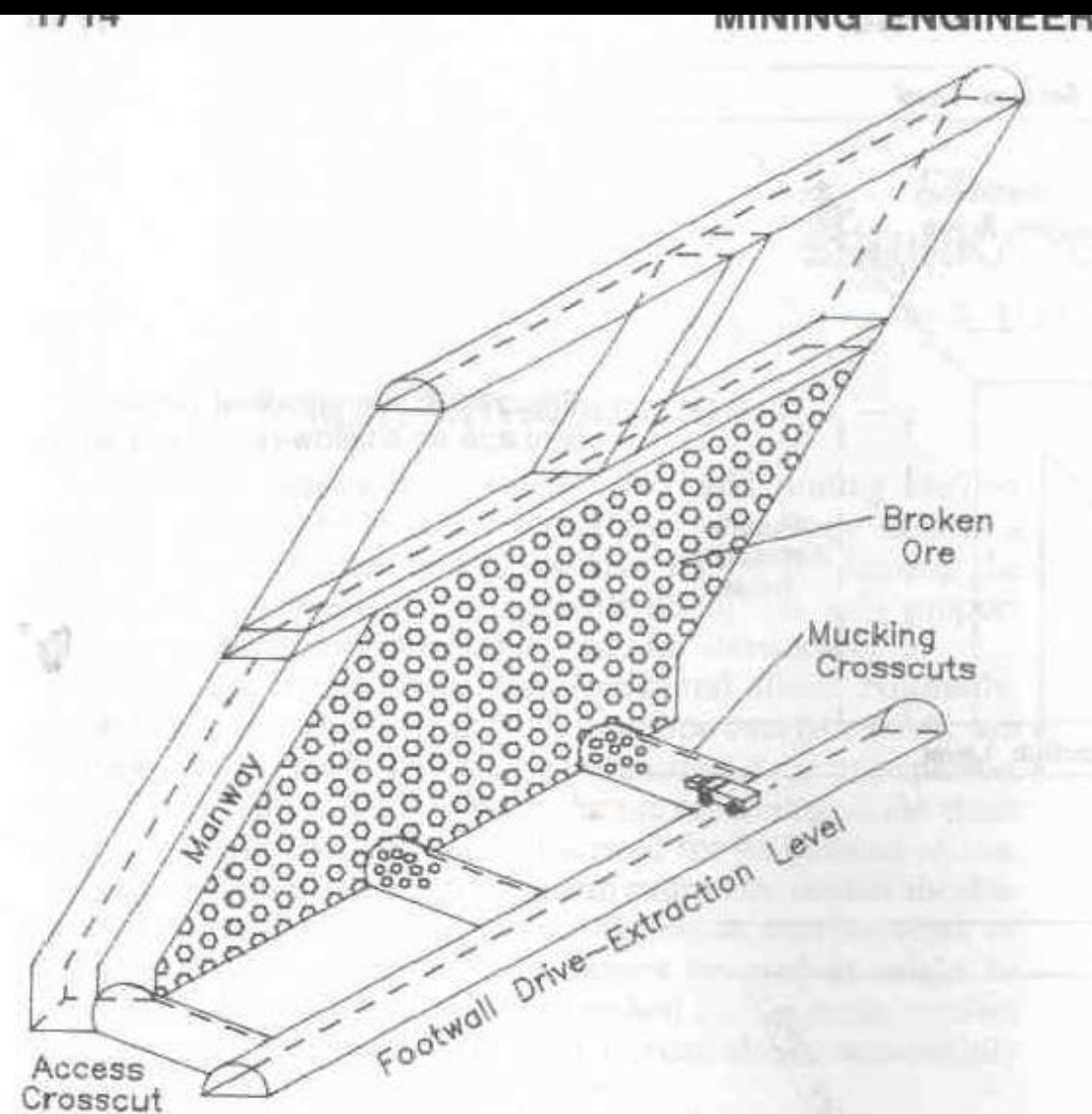


Fig. 18.3.4. Typical shrinkage stope with LHD extraction.

Stope Preparation 1

- *Raises are fixed with ladder for movement of persons.
- *Fixing of ladder in raise is generally done from upper level to lower level.
- *Men ways & service way are constructed on both side of stopes.
- *A timber slide is installed in one of man ways for hoisting & lowering material in to & out of stope.
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Stoping Preparation 2

*Hoisting of material is done by single drum air hoist installed at raise mouth.

*After establishing man ways, ventilation raise & service ways, stoping operation commences.

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Stoping 1

- *Upper hand stoping method is adopted.
- *Broken ore is allowed to be in stope.
- *In shrinkage stoping, broken ore remains in stope to form a working platform for worker in side stope.
- *Left out ore in stope also extend support to hang wall till extraction of ore from stopes tale place.

Stoping 2

- *Stoping takes place by horizontal slices with breast method.
- *Normally 35% broken ore is taken out from stope to compensate swelling of ore by blasting.
- *Drawing of ore causes “shrinking” of ore volume in side stope, thus method is named “Shrinkage Method”.

Drilling & Blasting 1

- *Upper holes & breast holes drilling, both are practiced.
- *Back of sill level is drilled either by hand held drill machine.
- *For wider stope drilling is done by drill wagon or stope jumbo.
- *Normally back of stope is blasted by upper hole, however in some cases breast stoping is also done.
- *Horizontal cut is taken by hand hold machines.
- *Drill hole length for back stoping is 1.8m to 2.4m in length.

Drilling & Blasting 2

- *Normally complete back is blasted at one time with upper hole drilled.
- *Breast holes of are drilled, 1.8m to 3m long for blasting.
- *In breast blasting, normally one blast is taken in a day.
- *Drill holes are loaded with ANFO, water gel, slurry explosive, emulsion explosive.
- *Initiation is commonly done by non electric method, but electric blasting is also practiced.

Ore Drawing from Stope 1

- *A slice in ore is cut at bottom of stope block.
- *35% of ore is drawn out from stope, 35% is normal swell factor of blasted ore.
- *Top of muck pile is leveled to facilitate drilling for next cut.
- *Leveling of muck is done by hand shovel in small stope.
- *For leveling muck pile, Slusher (scraper) or LHD is deployed in side large stope.

Ore Drawing from Stope 2

*After leveling of muck pile, drilling for next cut, raising of man way & so forth are done to continue mining cycle.

*While drawing ore from stope, some time, hang up (arches) form above draw chute.

*In this case, there is a possibility, that man standing in broken ore may get sucked up due to sudden collapse of arch.

Additional Service Opening in Stopping 1

*Some time timber cribbed opening, steel tube openings are required through broken ore.

*This installations are desirable during mining phase but may create safety problem & nuance, with collapse of material used to construct this opening.

*At time of construction of passages inside broken ore, this installation should be pinned or wedged to stope wall to prevent distraction of this structure at time of drawing ore from stope.

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Additional Service Opening in Stopping 2

*Destroyed man way be drowned down along with broken ore & clog chute gate.

*This material likely to cause hang up away from chute gate.

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Hang wall of Stope 1

- *Stope should have self supporting walls to permit application of shrinkage stoping.
- *Collapse from hang wall increases dilution in broken ore.
- *Collapse from hang wall increases at time of broken ore drawing from stop.
- *These increases dilution of ore in stope.
- *Heavy collapse even prevent stoping process by clogging ore flow at time of drawing broken ore.

Support 1

- *Hang wall to be supported while slicing ore from back.
- *It is better to support hang wall at time of slicing.
- *Support can be in situ rock pillar at random (waste patch) or systematic pillar.
- *Ore pillar left in stope can be drilled off & blasted while drawing broken ore from stope.
- *Timber support (stull) with plank head board should not be used as support to hang wall.
- *Timber subsequently gets buried in broken rock while mining upper slices.

Support 2

- *This timber becomes hazard at time of drawing down ore from stope.
- *This timber causes plugging at chute & hang up above chute.
- *Rock bolting is a preferred support for shrinkage stope.
- *Both mechanical rock bolt or grouted type rock bolt can be used for stope support.
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Support 3

*In narrow shrinkage stope, some time installing rock bolt at hang wall becomes difficult due to space constrained in stope.

*In this condition, all rock bolt may not be perpendicular to HW of stope.

Sampling 1

- *Sampling of narrow shrinkage stope is done at stope back by taking channel or chip sample by hand or through mechanized means.
- *Sampling is systematically done at regular interval of 1.5m along entire back (hang wall to foot wall).
- *In wide stope, drill sampling at back of stope, hang wall & foot wall is done.
- *Drill hole samples are collected from back, hang wall & footwall at perpendicular pattern.
- *Drill cutting sample are collected in a sample bag through a hose & funnel or any other device.

Stope Draw Down 1

- *Draw down ore from shrinkage stope is difficult & dangerous process.
- *Level of difficulty goes up, where ore contains sticky material thus causing hang up between stope wall.
- *Hang up are to be washed down with water or to be blasted down with explosive or re-mine broken ore.
- *Pick down by miner should be avoided as there is always possibility of miner getting buried under clogged ore.
- *In extreme condition, stope is abandoned along with broken ore.

Stope Draw Down 2

*Shrinkage stoping should not be done, where ore has tendency to hang up.

Draw Pattern 1

- *Ore from stope should be drawn systematically.
- *Drawing of ore should be done evenly, in a regular sequence.
- *In case of failure of stope wall, waste rock should remain above broken ore.
- *Broken ore should not get trapped in between waste rock.

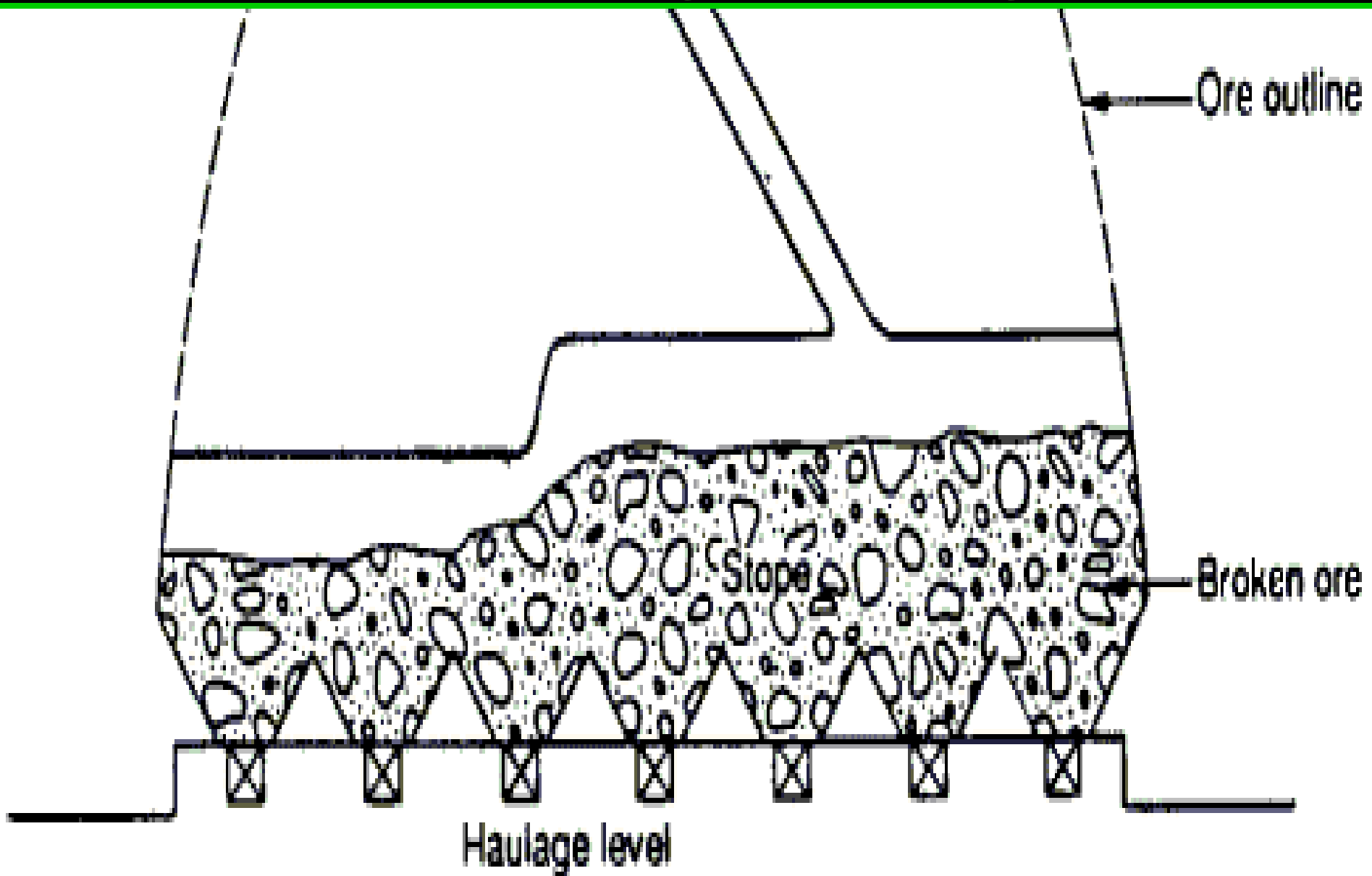
Draw Pattern 2

- *At time of drawing ore, operators control over walls, pillar, recovery etc. is minimal.
- *Re-entry of miner in to stope under active draw down should not be done at any condition.
- *It is very risky & miner may get buried under broken ore as ore is under movement.

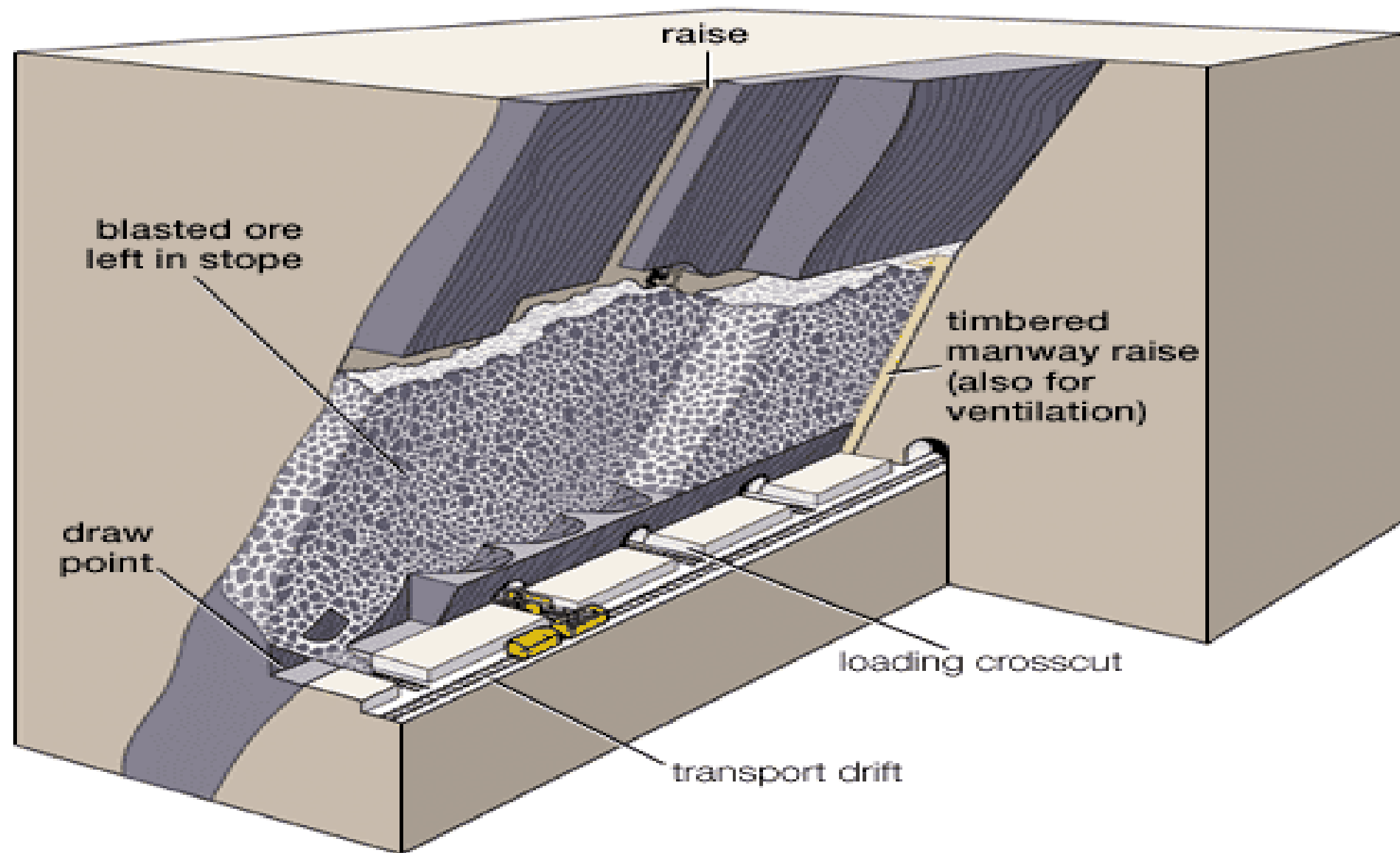
Draw Point 1

- *Stopes are to be drawn from strategically placed chute or draw point cross cut.
- *Haulage from stope extraction point can be done with track mounted equipment or LHD, mine truck.
- *Chute should be robustly designed and constructed, so that they does not get destroyed while blasting long slab in them.
- *Stope can be extracted through Slusher (scrapper) trench's developed below stope.

Shrinkage Stopping



Shrinkage Stopping



Long Hole Shrinkage 1

- *Long hole shrinkage stope is developed as normal shrinkage stope.
- *One raise in ore body, equipped with Alimak Raise climber is located at middle of stope.
- *Horizontal long holes are drilled in both direction along strike of ore body.
- *These long holes are blasted from Alimak raise climber.
- *Alimak raise climber serves for entry & exit to stope.
- *Length of hole is maximum 30m in either direction, totaling 60m strike length of stope.

Long Hole Shrinkage Stope

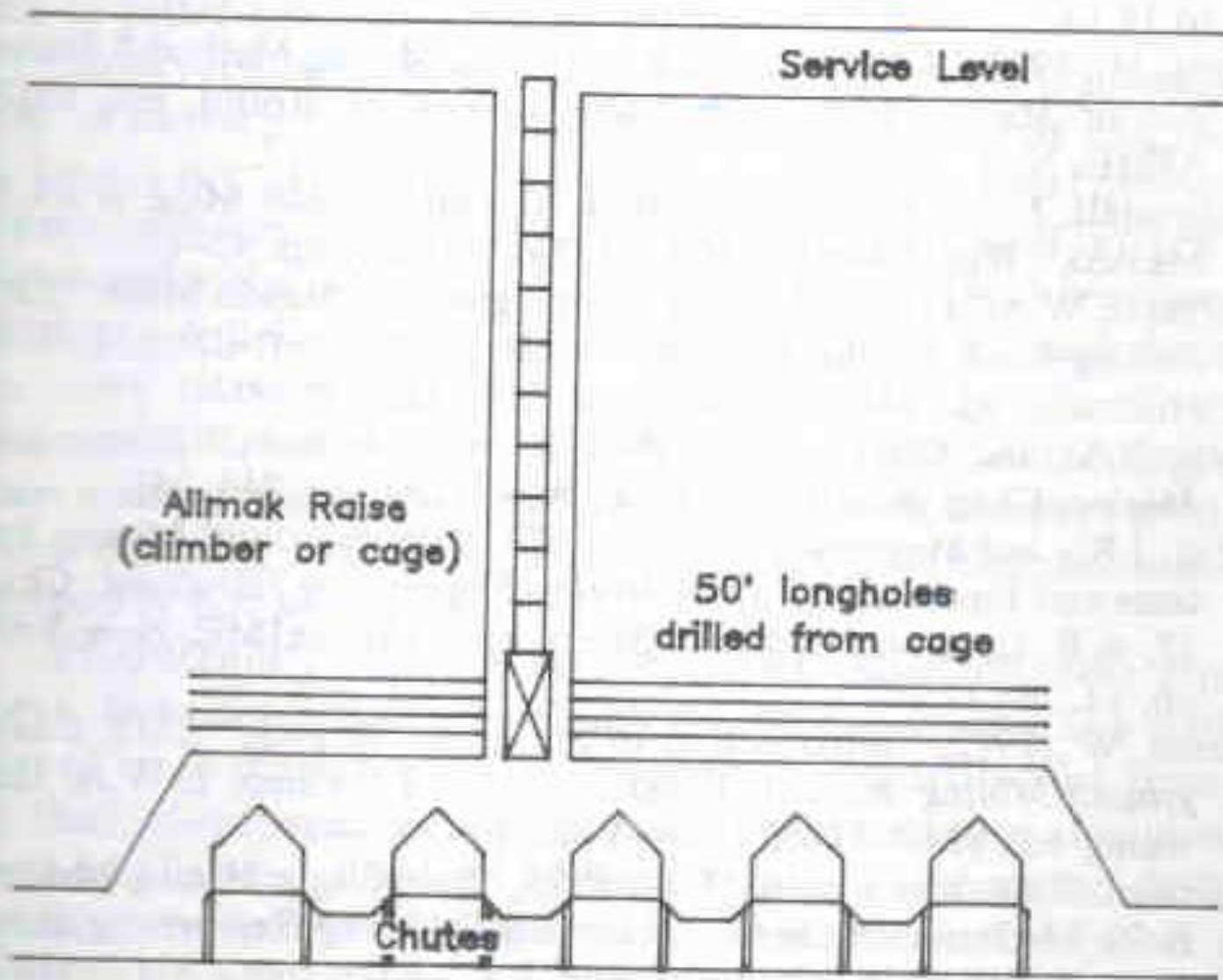


Fig. 18.3.6. Longhole shrinkage stoping.

Shrinkage for Construction 1

- *Large cross section winze, shaft, raise are excavated deploying shrinkage method.
- *This excavation are divided in at least two compartment.
- *Main compartment is kept filled with blasted broken rock & provides platform for working.
- *After each blast, some broken rock is withdrawn from bottom chute in controlled manner.
- *After completion of excavation, all broken rock are removed, compartments are dismantled.

Shrinkage Constriction

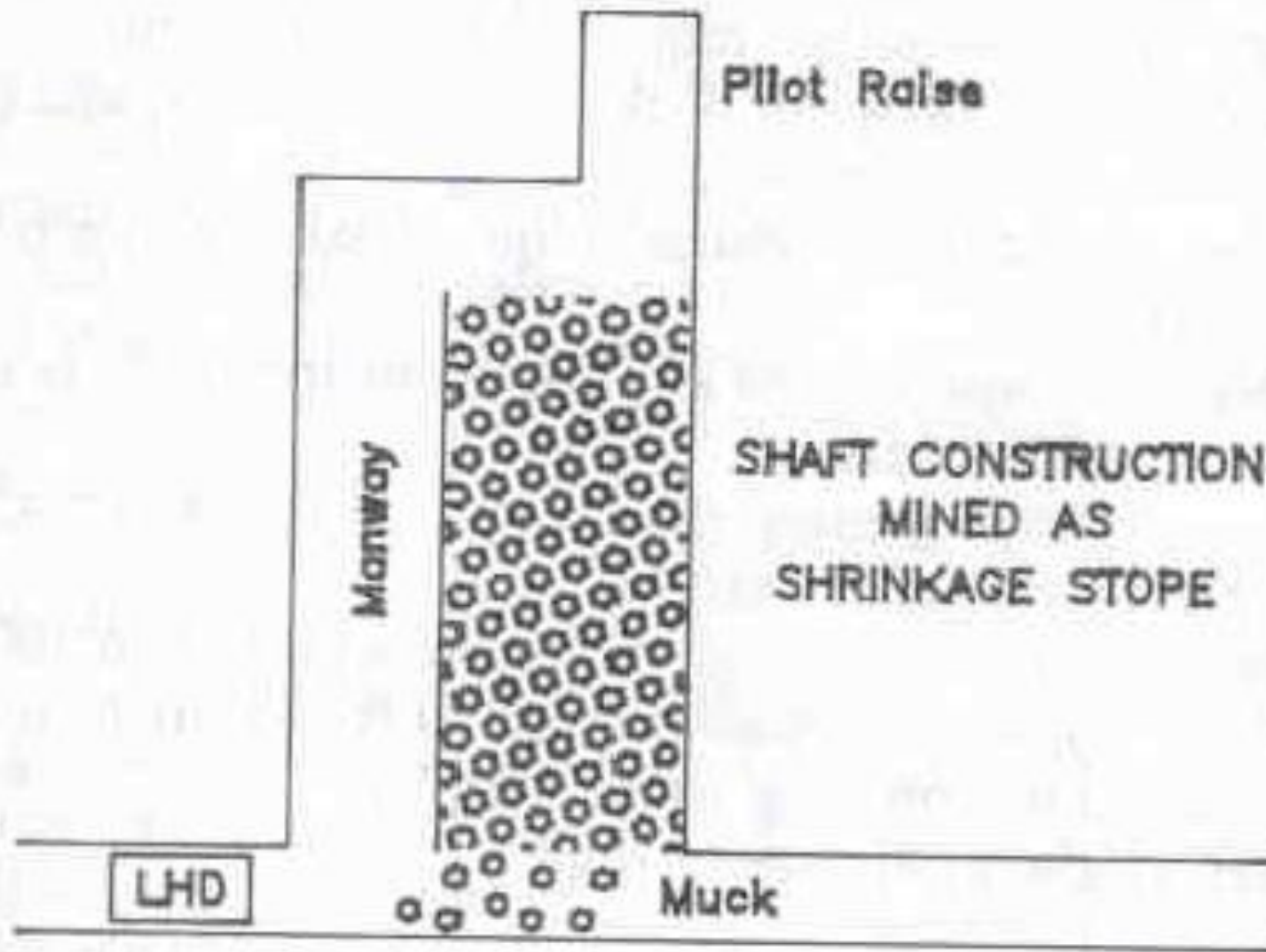


Fig. 18.3.7. Construction shrinkage stoping by conventional method.

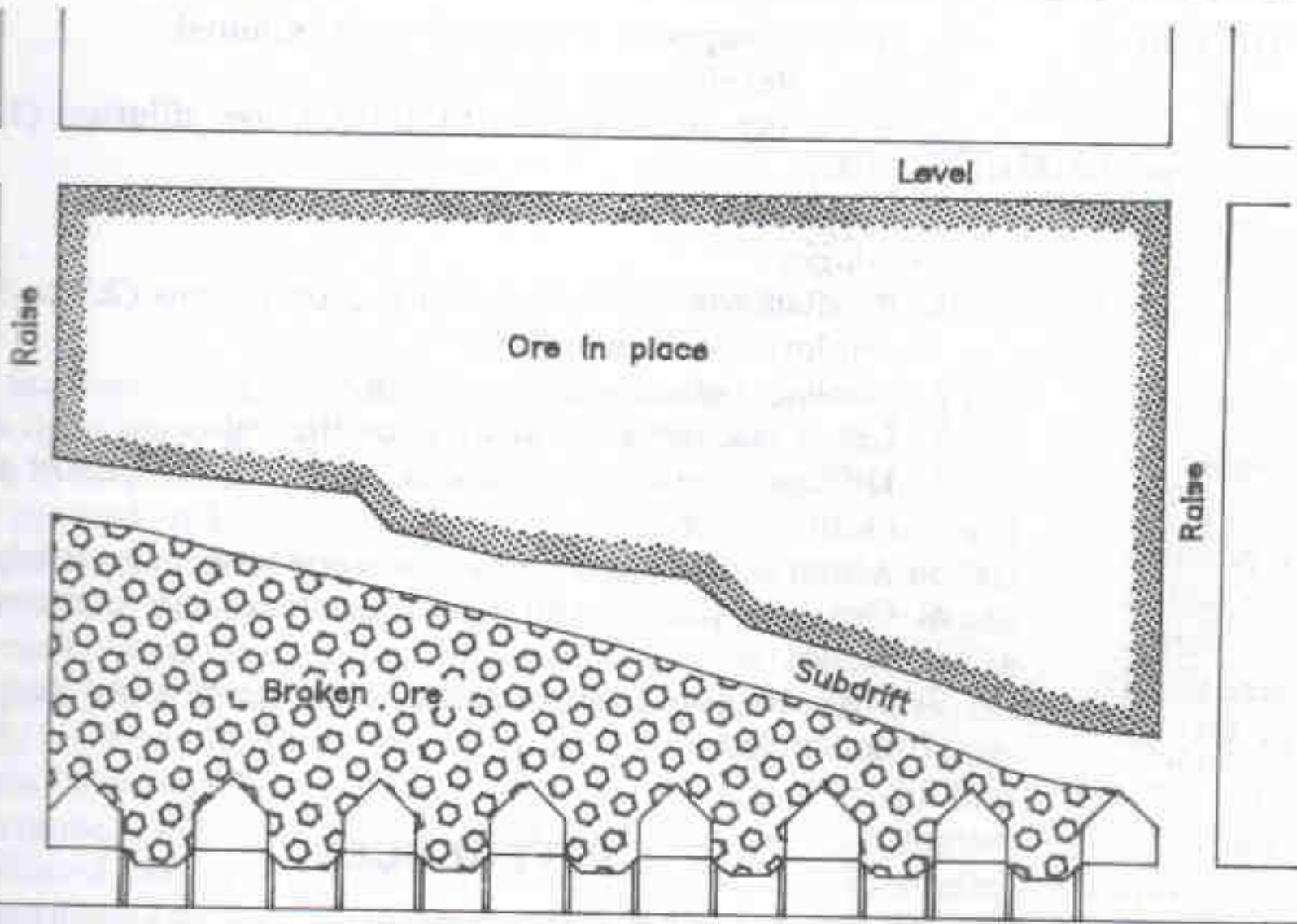
Incline Shrinkage Stope 1

- *Incline shrinkage is like rill stoping with multiple faces or benches.
- *Each face (bench) is drilled & blasted independently, thus increasing space of breakage in stope.
- *Breast drilling is practiced with pneumatic drill machine & air leg.

Shrinkage Rill Stope 1

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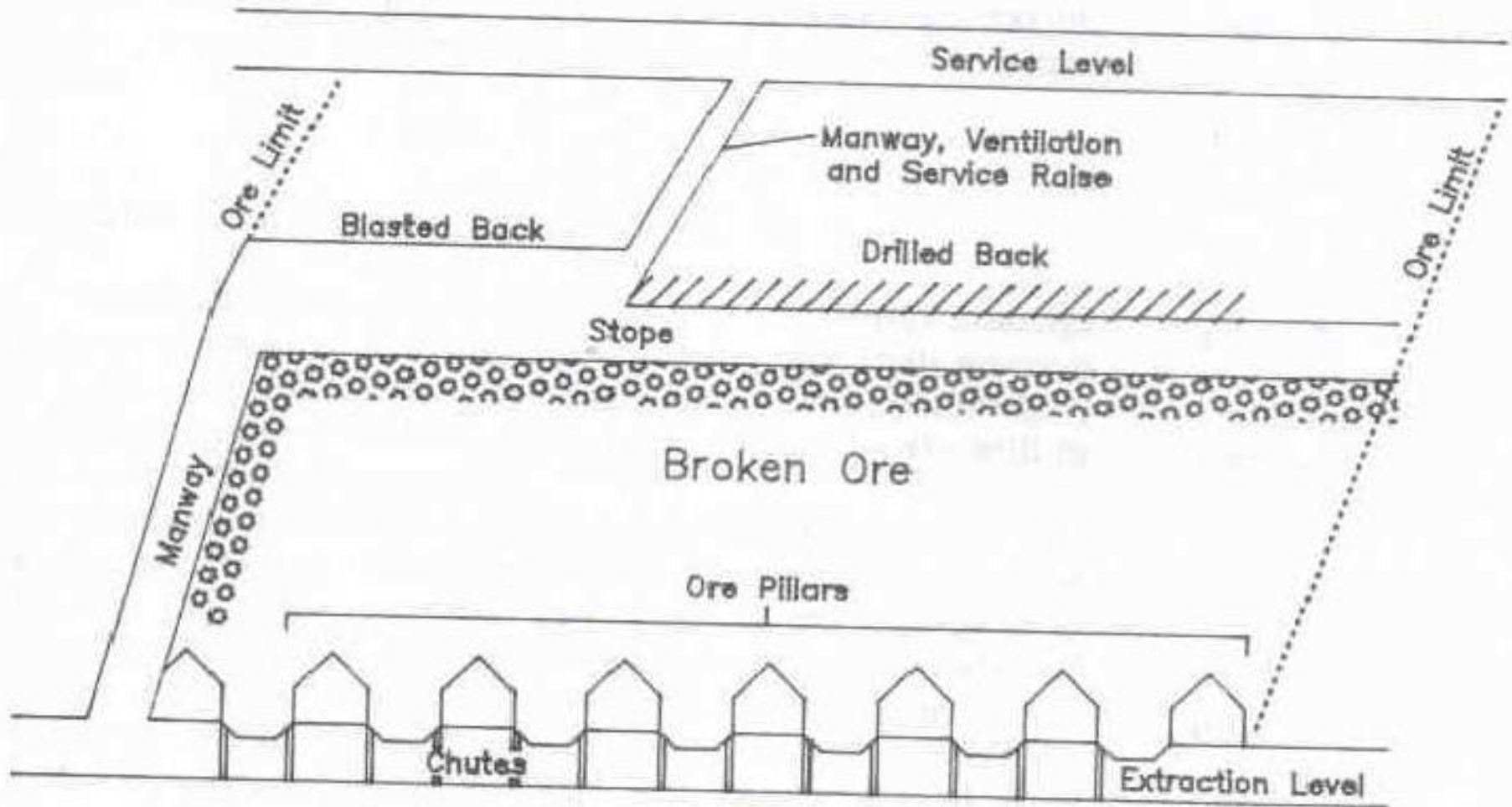
SHRINKAGE STOPPING



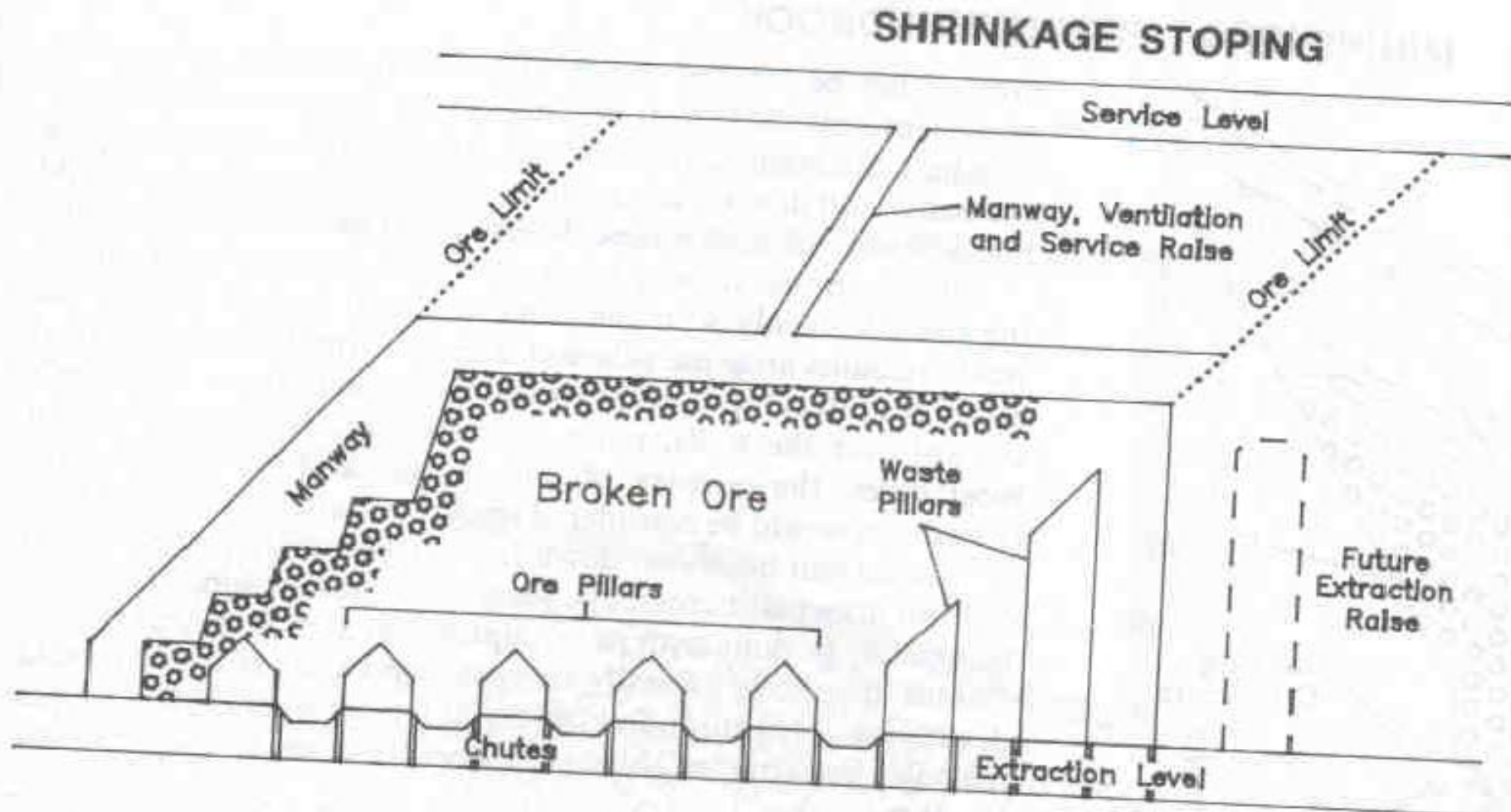
Small Ore Body 1

- *Small ore body as a hole, can be mined by shrinkage method with relatively small investment.
- *Small ore body with shallow plungers (rake less than 50 degree) will be difficult to mine by shrinkage method.
- *In this case ore moves away too quickly from pre developed extraction system.
- *Ore body abruptly extend for grate distance then additional development, specially additional raisings are required to extract ore from stope.

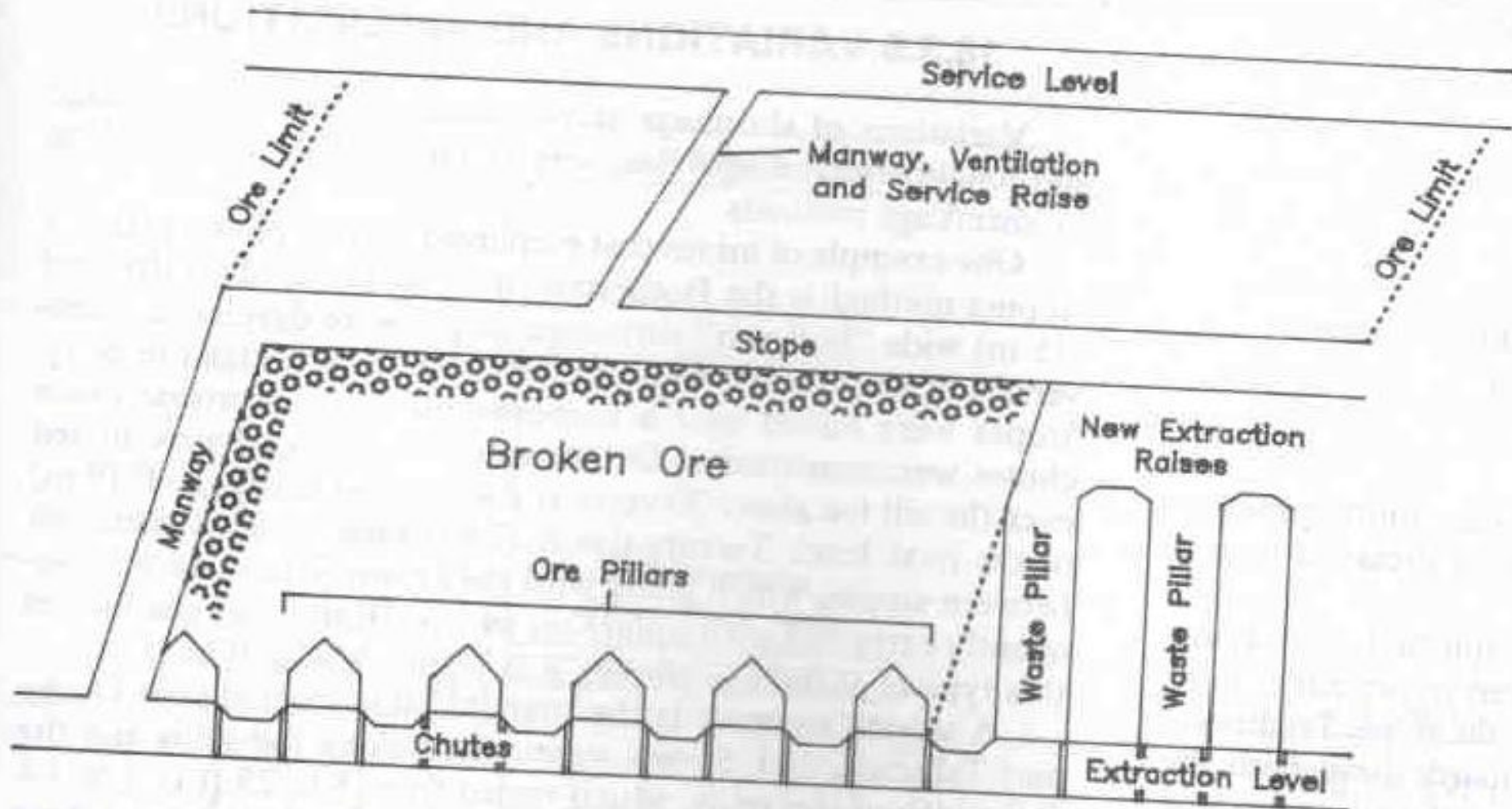
Typical Shrinkage Stope Longitudinal Section



Shrinkage Stope Shallow Raking



Shrinkage On Irregular Ore Body



Advantages 1

- *Broken rock gives support to HW of the stope.
- *Mining can be selective by leaving some waste pillars in stope.
- *Smaller capital out lay & less development required.
- *This method is applicable to narrow ore body or narrow vein, where other method can not be used.
- *This method can be applied to ore body from 1.2m to 30m wide.
- *Small ore body can be mined, due to low capital investment.

Advantages 2

- *Ore pill left inside stope can be recovered at time of drawing ore from stope.

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Disadvantages 1

- *Major portion of ore is recovered only after completion of the stopes.
- *High ore inventory is locked up for considerable time
- *Presence of sticky matter or oxidizers in the ore creates difficulty.
- *Void created at draw point may close up suddenly causing accident.
- *Difficult to control draw, when stope is in advanced stage.

Disadvantages 2

- *Dilution from HW.
- *This method is labour intensive.
- *High level of mechanization is not possible.
- *Dilution from HW.
- *This method is labor intensive & can not be mechanized.
- *High level of human exposure in freshly exposed, blasted active stope.
- *Low ore recovery from stope.
- *At some time, ore may get blocked in stope permanently & lost for ever.

Reference 2

*SME mining Engineering Hand Book, 2nd Edition,
Volume 2, page 1712 to 1716